Search-Concentration Audits for Learned-Dynamics Tree Planning

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Abstract

Tree search over learned dynamics is usually presented as extra planning computation. This paper audits a complementary failure mode: adaptive tree expansion can concentrate model queries on a locally optimistic error and turn a learned simulator into its own adversarial objective. We build a controlled two-dimensional dynamics audit with a known reward and transition bias pocket, then compare a static rollout pool, UCT MCTS, value-guided MCTS, and uncertainty-aware repairs under equal forward-model budgets. The main result is a tail-risk finding, not a dominance claim. At budget 1024, UCT MCTS has a larger mean selected-return optimism gap than the static pool (0.686 versus 0.301) and a larger worst branch-capture event (7.021), while the paired median delta is near zero and UCT exceeds the static pool on only 20% of seeds. An uncertainty-penalized search objective reduces the mean gap to 0.236 and the 90th-percentile gap to 0.156 in the same artifact. The contribution is a reproducible audit protocol for detecting when adaptive planning spends computation on model optimism rather than real return.

1 Introduction

Learned model-based agents such as PlaNet, Dreamer, and PETS show that learned dynamics can support efficient control (Hafner et al., 2019; 2020; Chua et al., 2018). Recent tree-search work further asks how to plan when transitions are uncertain or continuous (Kohankhaki et al., 2024; Riviere et al., 2024). In all of these systems, additional model queries are valuable only if the model remains a trustworthy evaluator over the region that search discovers.

This paper studies what happens when that premise fails locally. If the same learned dynamics and reward surrogate both proposes and scores futures, adaptive search becomes a test-time optimizer over a fallible simulator. A branch that accidentally enters an optimistic pocket can receive higher backed-up value, attract more visits, and receive still more model queries. The error is therefore not merely sampled; it is amplified by the search allocation rule.

We frame this as a search-concentration audit. The audit compares static non-adaptive rollout selection to several adaptive tree-search variants under identical forward-model budgets, and it records not only selected return but also model-return optimism, reward-bias exposure, transition error, root visit concentration, depth-wise bias, and paired seed deltas. This lets the paper ask a precise question: does adaptive search create rare but severe branch-capture events under a biased learned model?

The answer in this controlled artifact is yes, but in a scoped way. UCT MCTS has a higher mean optimism gap and a much larger worst event at the largest budget, yet it does not stochastically dominate the static rollout pool in paired seeds. This distinction matters. A broad claim that tree search is generally more optimistic would be false for these data; the supported claim is that adaptive allocation can create a heavier optimism tail, and that uncertainty-aware scoring can shrink that tail when uncertainty is aligned with the bias pocket.

2 Related Work

Model-based reinforcement learning uses learned transition models for planning, policy improvement, or imagined value learning. PETS combines probabilistic ensembles with model-predictive control (Chua et al., 2018), while PlaNet and Dreamer learn latent dynamics for planning or behavior learning through imagination (Hafner et al., 2019; 2020). These methods motivate test-time planning with learned models, but also make model bias a central evaluation risk.

MCTS is a canonical adaptive allocation procedure. Recent work studies tree search under transition uncertainty and continuous dynamical systems (Kohankhaki et al., 2024; Riviere et al., 2024). Our focus is different from improving expansion rules: we ask whether the allocation rule itself can concentrate computation on local model optimism.

Inference-time compute work also separates independent sampling from adaptive search against a learned scorer (Snell et al., 2024). We import that lens into learned-dynamics planning without claiming an algorithmic benchmark advance. The contribution is an audit that exposes when additional planning computation is spent on a biased evaluator.

3 Audit Design

3.1 Controlled Dynamics

The environment is a deterministic two-dimensional point robot. The true reward encourages progress to a goal, penalizes action cost, and mildly penalizes an off-route pocket. The surrogate dynamics and reward model is accurate almost everywhere but, near that pocket, predicts extra reward and a transition drift toward the goal. It also reports elevated uncertainty in the same region. This makes the mechanism observable without claiming that the surrogate was trained from data.

For a selected action sequence $a_{0:H-1}$, the primary audit metric is

$$\Delta_{\text{opt}} = \hat{G}(a_{0:H-1}) - G(a_{0:H-1}),$$

where $\hat{G}$ is the surrogate rollout return and G is the true rollout return for the same sequence. Lower Δ_{opt} is better. The audit also records cumulative reward bias, transition error, uncertainty exposure, root visit concentration, depth-wise reward bias, tail quantiles, and paired deltas against the static rollout pool.

3.2 Planners

All planners use the same discrete action set, horizon $H = 18$, discount $\gamma = 0.98$, and forward-model budget accounting. The suite includes a random open-loop baseline, a static rollout pool that scores independent open-loop trajectories once, UCT MCTS, value-guided MCTS, uncertainty-penalized MCTS, and conservative-backup MCTS. The static pool is a non-adaptive control: it can select an optimistic trajectory, but it cannot use earlier optimistic evaluations to allocate additional evaluations toward the same region. MCTS adaptively expands and backs up branches according to UCB-style selection.

The intended repair is a lower-confidence search target. If the uncertainty signal $u(s, a)$ tracks local optimism, replacing $\hat{r}$ with $\hat{r} - \lambda u$ during scoring or backup can suppress the optimistic branch before it dominates root statistics. This is a controlled calibration assumption rather than a general statistical guarantee.

4 Experiments

The full run uses 20 random seeds and five budget settings from 64 to 1024. The committed artifacts include per-plan metrics, closed-loop episode diagnostics at the largest budget, root statistics, tail summaries, paired deltas, and the figures used below. The headline result is recorded in `results/full/manifest.json`; mean diagnostics, tail metrics, and paired comparisons are stored in `results/full/summary.csv`, `results/full/tail_metrics.csv`, and `results/full/pairwise_deltas.csv`.

Table 1 shows the largest-budget slice. UCT MCTS has the highest mean optimism gap among the main methods and a much larger maximum gap than the static rollout pool. However, the median UCT gap is essentially zero. The result is therefore a tail-risk diagnosis: adaptive search occasionally locks onto the optimistic pocket, and those events dominate the mean.

Table 1: Selected-return optimism tail at budget 1024. Lower values are better. Median, 90th percentile, and maximum expose whether a mean result reflects typical behavior or rare branch-capture events.

Planner	Mean	Median	90th pct.	Max
Static rollout pool	0.301	0.011	0.589	3.301
UCT MCTS	0.686	0.000	0.972	7.021
Value MCTS	0.397	0.001	0.196	6.803
Uncertainty MCTS	0.236	0.000	0.156	3.986
Conservative MCTS	0.261	0.000	0.431	4.300

Table 2: Paired seed deltas against the static rollout pool at budget 1024. Positive means the adaptive method had a larger optimism gap on that seed.

Planner minus static pool	Mean	Median	Positive fraction	Max
UCT MCTS	0.386	-0.000	0.20	7.021
Value MCTS	0.096	-0.000	0.40	6.802
Uncertainty MCTS	-0.065	-0.001	0.30	0.685
Conservative MCTS	-0.040	-0.000	0.35	4.247

Table 2 prevents overclaiming. UCT MCTS does not have a larger optimism gap on most paired seeds; its mean delta comes from rare large events. The uncertainty-penalized planner is the strongest repair in this run: it lowers the mean gap, the upper tail, and the paired mean delta, although it is not a formal guarantee under arbitrary uncertainty miscalibration.

Figure 1 plots the optimism gap over budgets. Figure 2 shows that tree search increasingly concentrates visits at the root as budgets grow. Figure 3 gives a depth-wise view of reward bias along selected plans. Together, these diagnostics support the mechanism claim that adaptive search can allocate additional computation toward model-optimistic branches.

5 Reproducibility

The accompanying artifact is organized around relative paths only. The full experiment is produced by `experiments/run_mechanism.py` with mode `full` and output `results/full`; the smoke run changes the mode and output path to `smoke` and `results/smoke`. The experiments use deterministic seeds $0, \ldots, 19$ for the full run, and the generated CSV/JSON/PNG outputs are committed under `results/full/` and `paper/figures/`. The paper build is scripted by `scripts/build_paper.ps1`, which also exports the versioned Desktop PDF.

6 Discussion

The result supports a narrow mechanism claim: in this controlled model, adaptive tree search can produce rare branch-capture events where the selected surrogate return is much more optimistic than the true return under the same action sequence. The static rollout pool also sometimes selects optimistic trajectories, but it lacks a feedback loop that concentrates further model calls on a discovered optimistic branch.

The experiment should be read as a microscope, not a benchmark. Its value is that the failure mode is observable, testable, and easy to extend. It does not establish that the same effect dominates in high-dimensional robotics tasks, nor that the uncertainty penalty transfers across model classes. The paired statistics are included specifically to prevent the mechanism claim from being mistaken for a broad dominance claim.

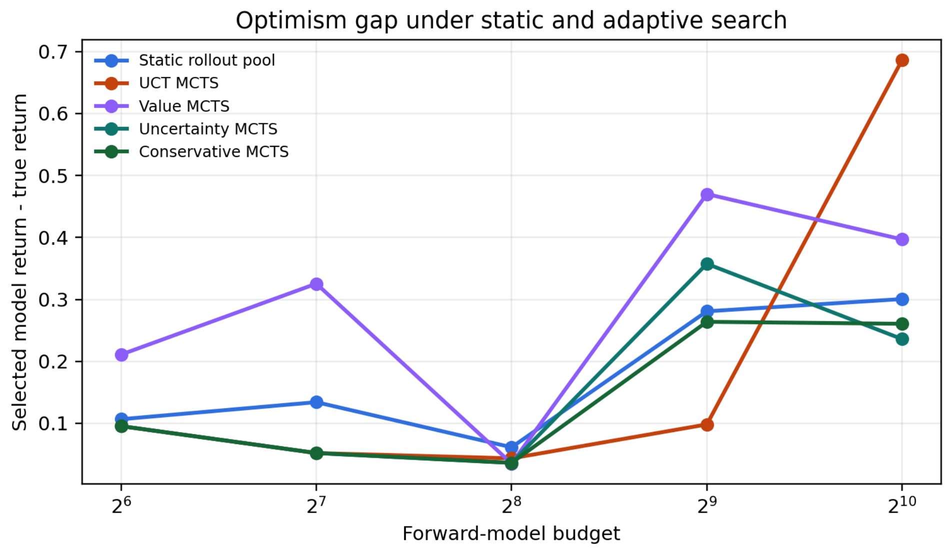

Figure 1: Mean selected-return optimism gap across forward-model budgets.

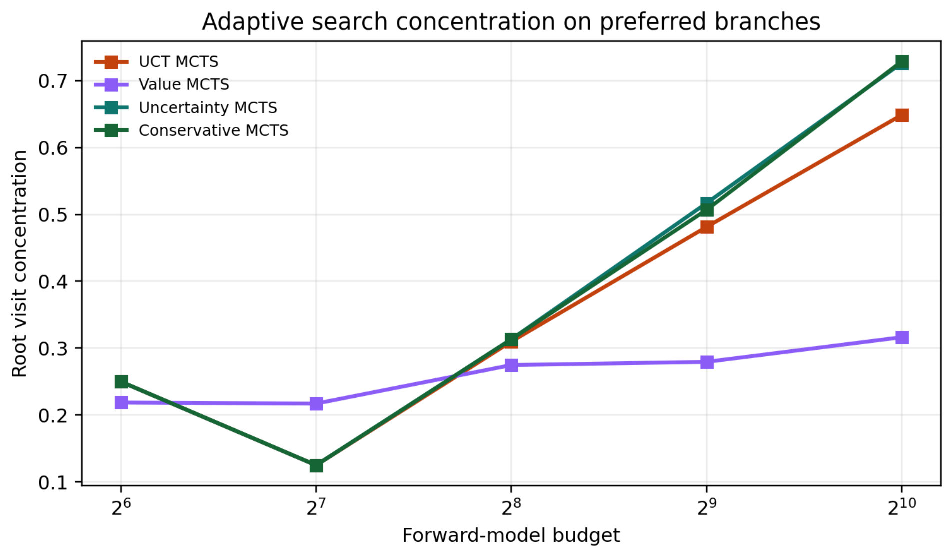

Figure 2: Root visit concentration as adaptive tree search allocates budget to preferred branches.

7 Limitations and Next Steps

The surrogate model is hand-designed rather than trained from data. The uncertainty signal is intentionally aligned with the injected bias pocket. The curves are seed-sensitive because the task is deliberately small, and no statistical claim beyond the committed repeated-seed run is made here.

The next version should train an ensemble dynamics model on biased or off-policy data, add continuous-action expansion or sampling-based proposals, and evaluate on at least one standard continuous-control benchmark. The repair should be stress-tested under miscalibrated uncertainty, value-function optimism, and biased action proposals. The most important open question is whether search concentration predicts real downstream regret when the model is learned rather than specified.

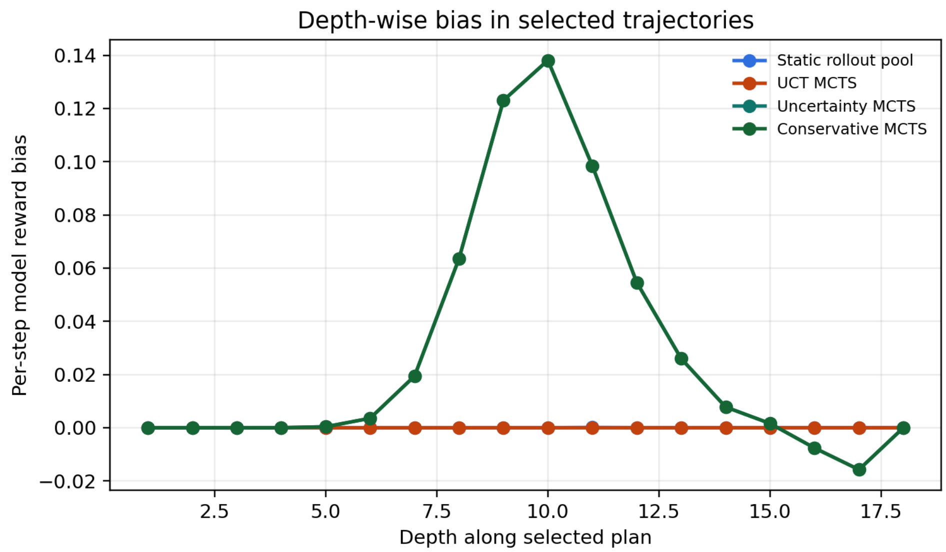

Figure 3: Depth-wise surrogate reward bias along selected plans in the largest-budget run.

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